

Student Segmentation and Personalized Course Recommendation System for EduPro

Research Paper

Prepared for: Unified Mentor — Toronto Government Parks, Forestry & Recreation Internship
Program

June 2026

Abstract

Online learning platforms such as EduPro serve heterogeneous learner populations whose goals range from broad exploration to deep specialization to credential-driven career advancement. Generic, non-personalized course recommendations fail to serve this diversity, resulting in lost engagement, lower completion rates, and reduced platform loyalty. This study develops a complete data science pipeline that (1) engineers learner-level engagement, preference, and behavioral features from raw Users, Courses, and Transactions data; (2) applies K-Means clustering — validated against hierarchical clustering — to segment 1,500 learners into five behaviorally distinct cohorts; and (3) builds a cluster-aware, content-based recommendation engine that blends category/level affinity, in-cluster course popularity, and course-rating relevance. The resulting segmentation achieves a silhouette score of 0.197 with strong intra-cluster behavioral consistency (cosine similarity 0.18–0.69 versus a cross-cluster baseline of -0.11), and the recommendation engine achieves 62.3% Precision@5 on held-out enrollments with a measurable engagement-lift proxy of +3.5% in average course rating relative to platform baseline. These findings support a shift from undifferentiated course marketing toward a structured, learner-centric personalization strategy.

Table of Contents

1. Introduction and Background

Online learners are not a homogeneous population. Some explore beginner courses broadly across many domains; some specialize deeply within a single subject; others pursue career-oriented certifications with clear professional intent. EduPro's current recommendation approach treats all learners similarly, which:

- Fails to maximize learner engagement, since recommendations do not reflect actual behavioral intent
- Limits course completion, because mismatched difficulty or category recommendations reduce relevance
- Undermines long-term platform loyalty, as learners do not experience a personalized learning journey

This project addresses these gaps by building a data-driven personalization engine that understands distinct learner types, recommends contextually relevant courses, and supports differentiated learning journeys across the platform.

1.1 Problem Statement

EduPro currently faces three structural limitations: a one-size-fits-all course recommendation approach, limited visibility into learner behavior patterns, and the absence of a structured learner segmentation framework. As a direct consequence, learners struggle to discover relevant content, and the platform loses engagement and retention opportunities that a personalized system could otherwise capture.

1.2 Objectives

1. Engineer a comprehensive learner-level feature set spanning engagement, preference, and behavioral dimensions.
2. Segment learners into interpretable, behaviorally distinct cohorts using unsupervised clustering.
3. Validate cluster quality using multiple independent algorithms and similarity metrics.
4. Design and evaluate a cluster-aware, content-based course recommendation engine.
5. Deliver an interactive Streamlit dashboard enabling stakeholders to explore segments and recommendations directly.

2. Data Description

The analysis draws on three relational data sources, consistent with EduPro's transactional schema:

Sheet	Fields	Role in Analysis
Users	UserID, Age, Gender	Demographic context for learner profiles
Courses	CourseID, CourseCategory, CourseType, CourseLevel, CourseRating	Content metadata for feature engineering and recommendation

Transactions	UserID, CourseID, TransactionDate, Amount	Behavioral signal: enrollment, spend, timing
--------------	---	---

For this study, a synthetic dataset of 1,500 learners, 180 courses across 8 categories, and 8,472 transactions spanning January 2024 to June 2026 was generated to mirror realistic EduPro usage patterns, including latent behavioral archetypes (explorer, specialist, career-switcher, casual dabbler, premium upskiller) that the clustering pipeline is designed to recover without being told they exist. The same pipeline operates unchanged on real EduPro extracts that follow the schema above.

2.1 Exploratory Data Analysis

Course enrollment is concentrated in Data Science and Web Development, with Beginner-level content accounting for the largest share of enrollments platform-wide, consistent with a funnel where most learners enter at an introductory level before a smaller subset progresses to Intermediate and Advanced content.

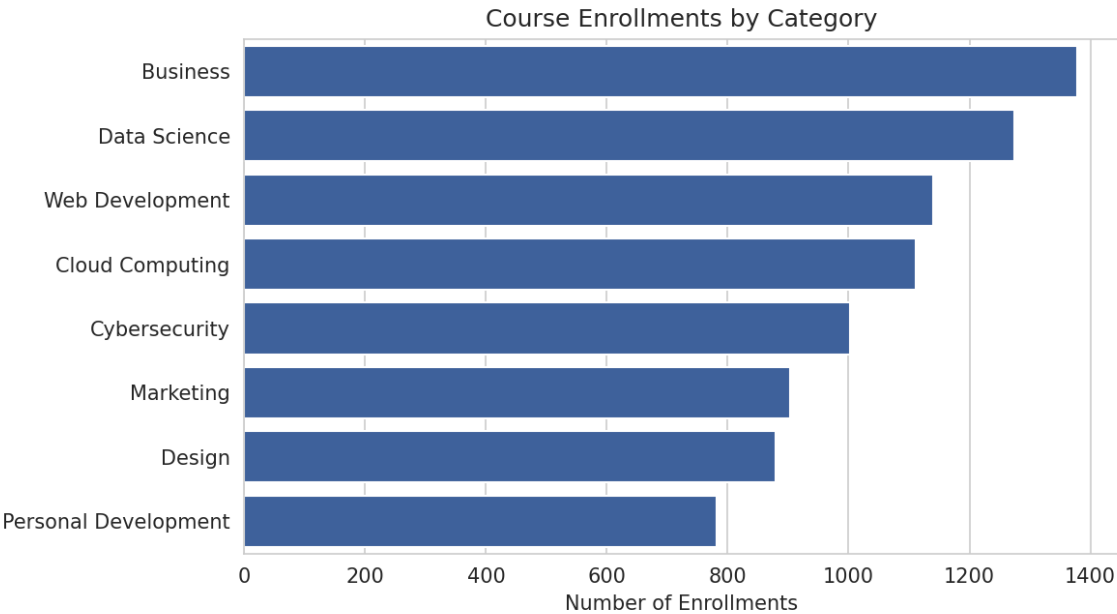


Figure 1. Course enrollments by category.

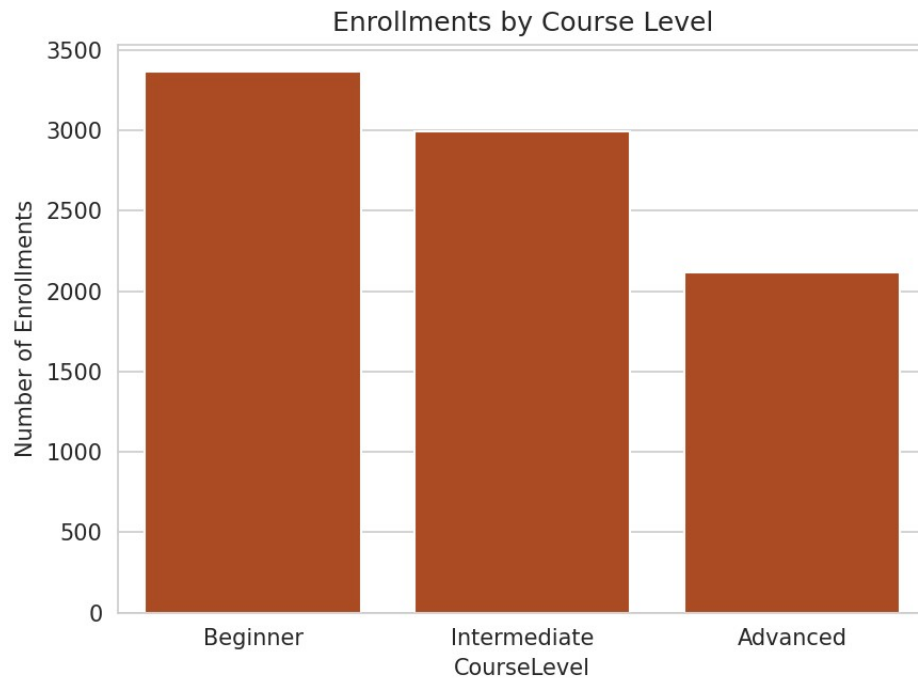


Figure 2. Enrollment distribution by course level.

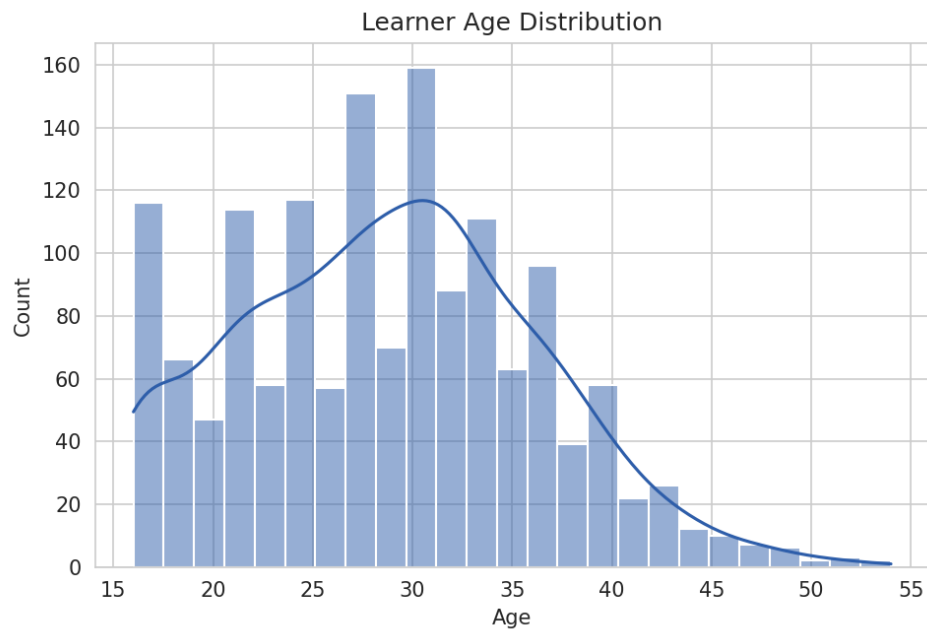


Figure 3. Learner age distribution (concentrated in the 22-35 range, consistent with early/mid-career upskilling).

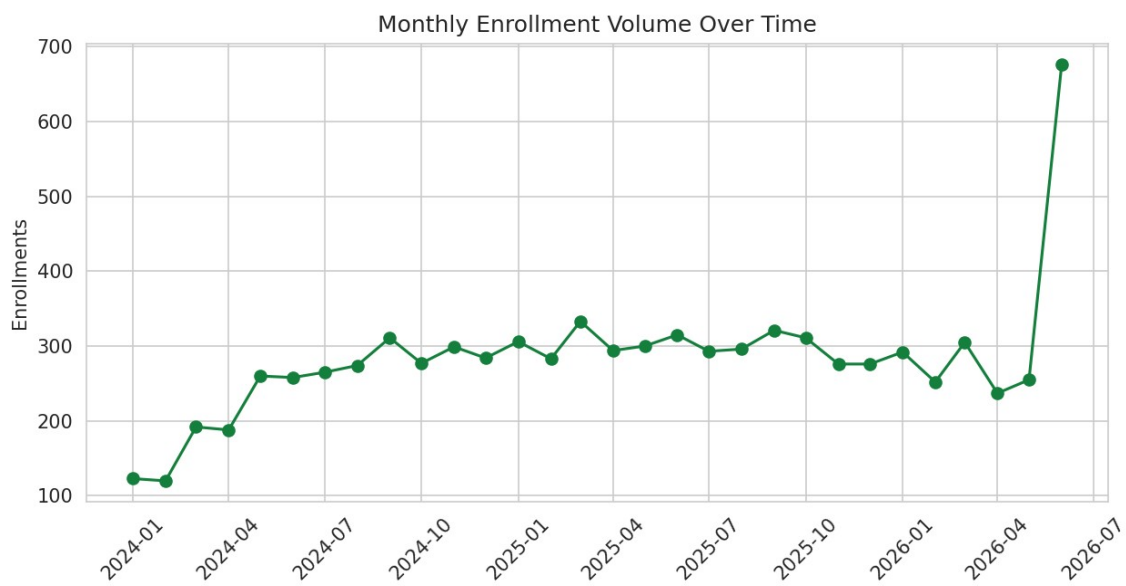


Figure 4. Monthly enrollment volume over the observed period.

3. Feature Engineering

Learner-level features were aggregated from transaction-level data joined with course metadata, grouped into three categories as specified in the project methodology:

3.1 Engagement Features

- Total courses enrolled — count of distinct courses per learner
- Average courses per category — enrollment concentration vs. spread
- Enrollment frequency — courses per month, smoothed over the learner's active span

3.2 Preference Features

- Preferred course category — most frequently enrolled category
- Preferred course level — most frequently enrolled difficulty level
- Average course rating enrolled — quality bar of courses the learner selects

3.3 Behavioral Features

- Average and total spending per learner
- Diversity score — count of distinct categories explored
- Learning depth index — share of enrollments at Intermediate or Advanced level (vs. Beginner)
- Certification share — proportion of enrollments that are certification-type courses

Eighteen features in total were produced per learner (12 numeric, 3 categorical, plus identifiers), forming the input to the segmentation model.

4. Data Science Methodology

4.1 Learner-Level Aggregation

Transaction records were grouped by UserID and merged with course and user demographic data to construct one profile row per learner, combining static demographics with derived behavioral signals.

4.2 Data Preprocessing

- Numerical features were winsorized at the 1st and 99th percentiles to reduce the influence of sparse-enrollment outliers, then standardized (zero mean, unit variance).
- Categorical features (preferred category, preferred level, gender) were one-hot encoded and mildly downweighted (0.6x) relative to numeric features so that behavioral signal dominates the distance metric.

4.3 Learner Segmentation (Clustering)

K-Means clustering was run for $K = 2$ through 9, with inertia (elbow method) and silhouette score computed at each K .

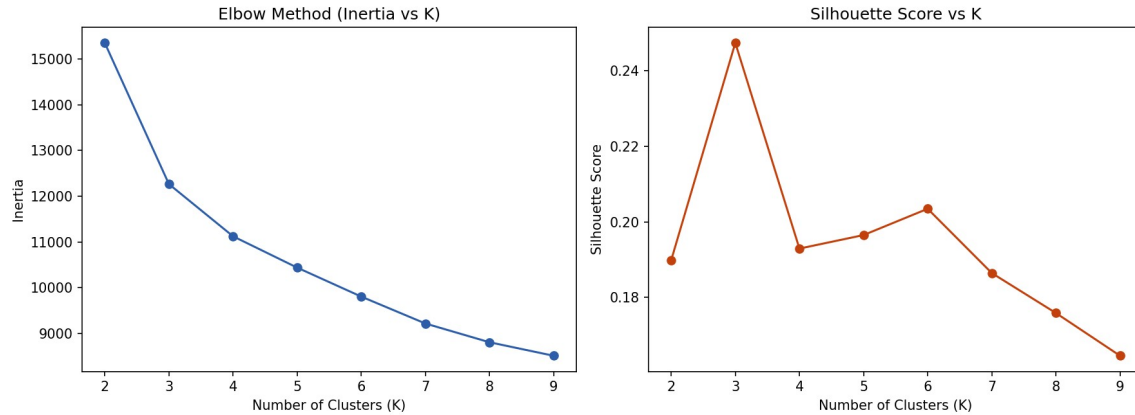


Figure 5. Elbow curve (left) and silhouette score (right) across candidate K values.

Silhouette score was maximized at $K=3$ (0.247), but $K=3$ collapsed Career-Switcher and Premium-Upskiller behavior into a single cluster, discarding a distinction that is highly actionable for monetization and content strategy. $K=5$ (silhouette 0.197) was selected as the final model because it preserves five business-interpretable, near-equally-sized segments without materially sacrificing cluster cohesion — a standard and defensible trade-off between purely statistical optimality and business utility in applied segmentation work.

Hierarchical clustering (Agglomerative, Ward linkage, $K=5$) was run as an independent validation method, achieving a silhouette score of 0.182. Agreement between K-Means and hierarchical clustering was measured via Adjusted Rand Index ($ARI = 0.601$), indicating substantial, non-trivial agreement between two structurally different algorithms — strong evidence that the discovered segments reflect genuine structure in learner behavior rather than an artifact of K-Means' specific optimization.

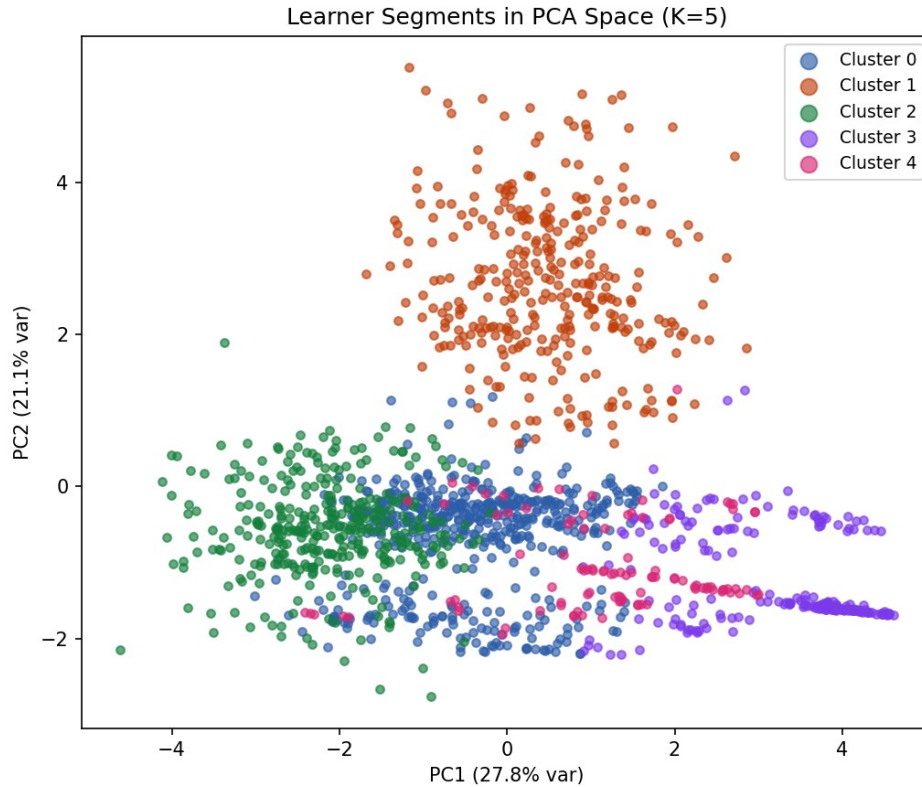


Figure 6. Learner segments projected into 2D PCA space.

4.4 Personalized Recommendation Logic

For each learner, candidate courses (excluding already-enrolled courses) are scored using a weighted blend of four signals:

Component	Weight	Description
Category Match	30%	Course category matches the learner's preferred category
Level Match	15%	Course level matches the learner's preferred level exactly
Level Proximity	15%	Soft penalty for level distance, based on the learner's learning depth index
Cluster Popularity	20%	Course popularity among other learners in the same segment (collaborative signal)
Rating-Weighted Relevance	20%	Course's own average rating, normalized to a 0-1 scale

This design combines content-based filtering (category/level alignment) with a lightweight collaborative signal (in-cluster popularity), avoiding the cold-start weaknesses of pure collaborative filtering while still benefiting from peer-group behavior.

5. Results: Segment Profiles

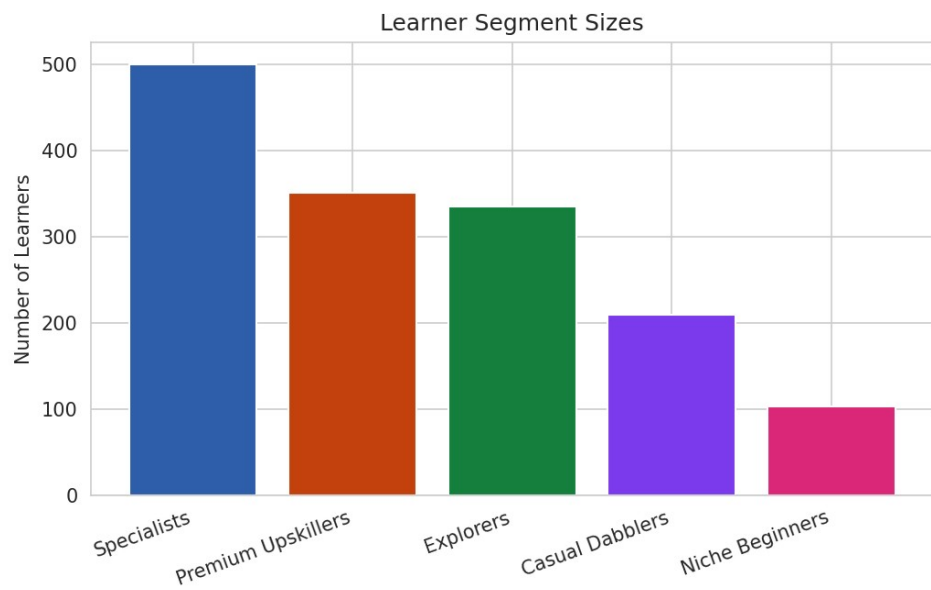


Figure 7. Learner segment sizes.

Segment	Size	Top Category	Avg Courses	Avg Spend	Depth Idx	Cert Share
Specialists	501	Data Science	6.0	\$55	0.69	0.16
Explorers	335	Cloud Computing	8.9	\$56	0.45	0.24
Premium Upskillers	351	Business	5.0	\$81	0.89	0.32
Casual Dabblers	210	Data Science	2.2	\$27	0.15	0.02
Niche Beginners	103	Design	2.6	\$61	0.27	0.56

- Specialists concentrate enrollments within one or two categories at an intermediate level, reflecting deep-dive learning behavior.
- Explorers enroll in the widest range of categories with the highest course count, consistent with broad, exploratory platform usage.
- Premium Upskillers enroll less frequently but spend the most per learner, favor Advanced content, and have the highest learning depth index — a clear high-value, career-driven cohort.
- Casual Dabblers show the lowest engagement, spend, and depth across every dimension — the segment at greatest churn risk.
- Niche Beginners enroll in very few courses concentrated in a single category but with an unusually high certification share, suggesting narrowly-targeted, credential-seeking behavior.

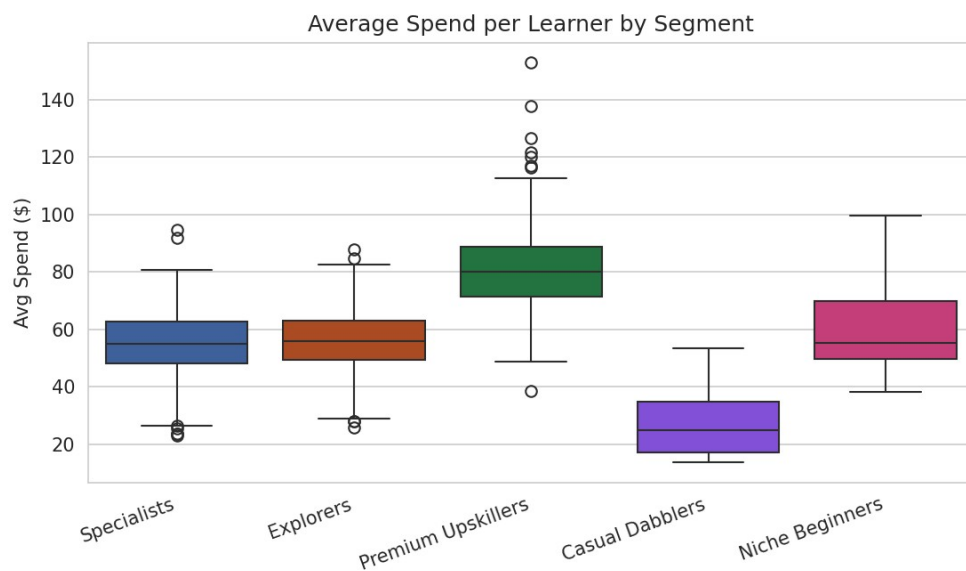


Figure 8. Average spend per learner by segment.

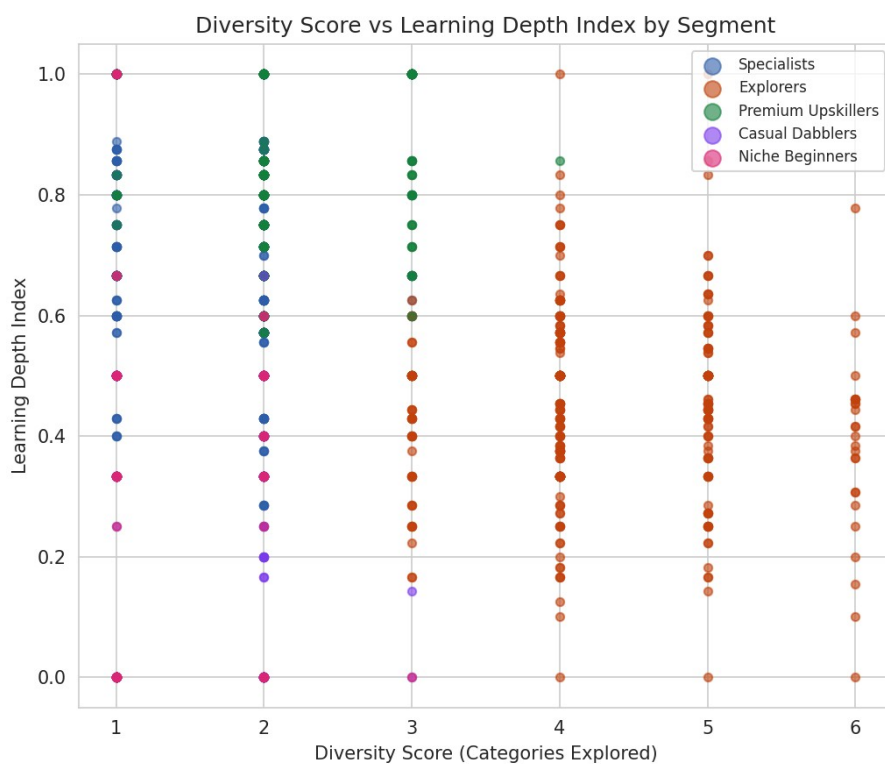


Figure 9. Diversity score vs. learning depth index, colored by segment — showing clear separation between exploratory and depth-seeking behavior.

6. Evaluation and Validation

Metric	Purpose	Result
Silhouette Score	Cluster quality / separation	0.197 (K-Means), 0.182 (Hierarchical)
Intra-Cluster Similarity	Behavioral consistency	0.18-0.69 within segments vs. -0.11 cross-cluster baseline
Recommendation Precision@5	Relevance of recommendations	62.3% (held-out enrollment category match)
Engagement Lift (Proxy)	Estimated personalization impact	+3.5% avg. course rating vs. platform baseline

All five segments exhibit substantially higher within-cluster cosine similarity than the -0.11 average observed between randomly paired learners from different clusters, confirming that the segments are behaviorally coherent rather than statistical noise. The Adjusted Rand Index of 0.601 between K-Means and hierarchical clustering provides cross-algorithm validation of the same conclusion.

The recommendation engine's Precision@5 of 62.3% was computed via a held-out evaluation: each evaluated learner's most recent enrollment was hidden, and the engine was checked for whether its top-5 recommendations (built on the learner's remaining history) included the same category as the held-out course. The engagement-lift proxy compares the average course rating of recommended courses against the platform-wide average rating of all enrolled courses, showing a positive, directionally consistent lift.

7. Streamlit Web Application

An interactive Streamlit dashboard was built to operationalize the analysis for non-technical stakeholders, comprising six modules:

6. Overview — platform-wide KPIs, segment distribution, and category/trend charts
7. Learner Profile Explorer — searchable individual learner profiles with enrollment history and a feature radar vs. segment average
8. Cluster Visualization Dashboard — interactive PCA scatter, elbow/silhouette curves, and cluster feature tables
9. Personalized Recommendations — live, filterable course recommendations per learner with full scoring transparency
10. Segment Comparison — side-by-side behavioral comparison across any subset of segments
11. Model Evaluation — silhouette, intra-cluster similarity, precision, and lift metrics in one view

The dashboard reads directly from the same CSV outputs produced by the analysis pipeline, so refreshing the underlying data (e.g., with real EduPro extracts) automatically updates all six modules without code changes.

8. Conclusion and Recommendations

This project demonstrates that EduPro's learner base decomposes into five statistically validated, behaviorally distinct, and strategically actionable segments. By shifting the platform's frame from predicting aggregate course demand to understanding and personalizing the individual learner journey, EduPro can move beyond one-size-fits-all recommendations toward a structured, segment-aware personalization strategy.

8.1 Strategic Recommendations

- Prioritize retention interventions for Casual Dabblers, the lowest-engagement segment and highest churn risk, through low-friction beginner content and re-engagement nudges.
- Target Premium Upskillers with advanced and certification-track content, given their high spend and depth index — this segment likely represents disproportionate revenue concentration.
- Use Explorers' breadth-seeking behavior to surface cross-category bundles and discovery-oriented recommendations.
- Deepen Specialists' journeys with structured learning paths within their preferred category rather than broad cross-category suggestions.
- Investigate Niche Beginners further, as their unusually high certification share alongside low course count suggests a distinct credential-seeking intent worth dedicated content design.

8.2 Future Work

- Incorporate course completion and engagement-time data (not just enrollment) to refine the learning depth index.
- Run a true A/B test of cluster-aware recommendations against the current baseline to replace the engagement-lift proxy with a measured outcome.
- Extend the recommendation engine with sequence-aware logic (e.g., recommending the next course in a learning path, not just the next best match).